



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

AIOT

03-04-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	A	C	A	D	A	C	C	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	C	D	B	C	B	A	D	B

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	D	D	B	B	C	D	B	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	A	A	D	A	B	B	C	B	A

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	C	B	B	B	A	D	B	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	D	D	C	A	D	D	C	C

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

2. Ans (C)

$$K_1 A \tan 30^\circ = K_2 A \tan 45^\circ = K_3 A \tan 60^\circ$$

$$\Rightarrow K_1 \tan 30^\circ = K_2 \tan 45^\circ = K_3 \tan 60^\circ$$

$$\Rightarrow \frac{K_1}{3\sqrt{3}} = \frac{K_2}{3} = \frac{K_3}{\sqrt{3}} \Rightarrow \frac{K_1}{3} = \frac{K_2}{\sqrt{3}} = \frac{K_3}{1}$$

3. Ans (A)

$$Mg(2R) + Mg(4R) = \frac{1}{2} \left[\frac{3}{2} MR^2 + M(2R)^2 \right] \omega^2$$

$$6MgR = \frac{11}{4} MR^2 \omega^2 \therefore \omega = \sqrt{\frac{24g}{11}}$$

4. Ans (C)

$$PT = \text{const} \quad PV = nRT, \quad T = \left(\frac{PV}{nR} \right)$$

$$\frac{P^2 V}{nR} = \text{const} \quad C = \frac{3R}{2} + 2R = \frac{7R}{2}$$

$$P^2 V = k' \quad C = 3.5 R$$

$$PV^{1/2} = k$$

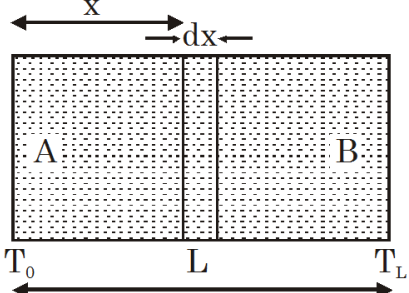
$$C = C_v + \frac{R}{1-n}$$

$$C = \frac{3}{2} R + \frac{R}{1-\frac{1}{2}}$$

5. **Ans (A)**

$$v = \sqrt{\frac{\gamma RT}{M}} = \sqrt{\frac{5RT}{3M}}$$

$$dx = C \cdot dt = \sqrt{\frac{5R}{3M}} \left[T_0 + \left(\frac{T_L - T_0}{L} \right) x \right] dt,$$

$$t = \frac{2L}{\sqrt{T_L} + \sqrt{T_0}} \sqrt{\frac{3M}{5R}}$$


6. **Ans (D)**

Given $y_1 = 4 \sin 600\pi t$, $y_2 = 5 \sin 608\pi t$
 $\therefore \omega_1 = 600\pi$ or $2\pi\nu_1 = 600\pi$ or $\nu_1 = 300$
 $A_1 = 4$
 and $\omega_2 = 608\pi$ or $2\pi\nu_2 = 608\pi$ or $\nu_2 = 304$
 $A_2 = 5$
 Number of beats heard per second
 $= \nu_2 - \nu_1 = 304 - 300 = 4$

$$\frac{I_{\max}}{I_{\min}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2} = \frac{(4 + 5)^2}{(4 - 5)^2} = \frac{81}{1}$$

7. **Ans (A)**

Let natural radius of loop is R and at any instant radius is r then mechanical energy of loop is

$$\frac{1}{2} K (2\pi r - 2\pi R)^2 + \frac{1}{2} M v^2 = \text{constant}$$

$$K 2\pi (r - R) 2\pi + \frac{M v dv}{dr} = 0$$

$$a = -4\pi^2 \frac{K}{M} (r - R) \Rightarrow T = \sqrt{\frac{M}{K}}$$

8. **Ans (C)**

$$C = \frac{\epsilon_0 A}{L} \therefore \log C = \log \epsilon_0 + \log A - \log L$$

$$\frac{dC}{C} = \frac{dA}{A} - \frac{dL}{L}$$

$$\Rightarrow \frac{dC}{C} = 2\alpha_1 dT - \alpha_2 dT = 0$$

$$\therefore 2\alpha_1 = \alpha_2$$

9. **Ans (C)**

Total power in horizontal resistors at any time

$$= i^2 R + \frac{i^2}{4} R + \frac{i^2}{16} R + \dots \dots \dots \infty = \frac{4}{3} i^2 R$$

Total power in vertical resistors

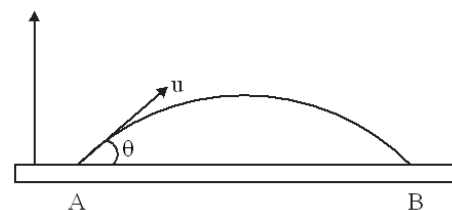
$$= \frac{i^2}{4} 2R + \frac{i^2}{16} 2R + \dots \dots \dots \infty = \frac{2}{3} i^2 R$$

$$\frac{P_{\text{Horizontal}}}{P_{\text{Vertical}}} = 2$$

10. **Ans (A)**

Electric field due to the large metal plate

$$E = \frac{\sigma}{\epsilon_0}$$



The projected electron's motion will be on a curved path like a projectile

$\therefore AB$ will be maximum when $\theta = 45^\circ$

$$\text{Range } AB_{\max} = \frac{u^2}{a}$$

$$a = \frac{\rho E}{m} = \frac{\rho \sigma}{\epsilon_0 m} = \frac{u^2}{\left(\frac{\rho \sigma}{\epsilon_0 m} \right)} = \frac{\epsilon_0 m u^2}{\rho \sigma}$$

11. **Ans (B)**

$$A_n = \pi r^2 \Rightarrow A_n = \pi (r_n n^2)^2$$

$$A_n = \pi r_0^2 n^4 \quad \text{so, } \ln \frac{A_n}{A_1} = 4 \ln n$$

12. **Ans (A)**

$$T(R - r) = \mu mg R, \quad 2mg - T = 2ma,$$

$$T - \mu mg = ma$$

On solving, we get

$$\therefore r = R \left(1 - \frac{3\mu}{2(1 + \mu)} \right)$$

13. **Ans (C)**

$$I_T = I_C \sqrt{1 + \frac{m^2}{2}} \Rightarrow I_C = \frac{11}{\sqrt{1 + \frac{(0.5)^2}{2}}} = 10.3$$

14. **Ans (D)**

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \rightarrow \text{differentiating twice w.r.t. time}$$

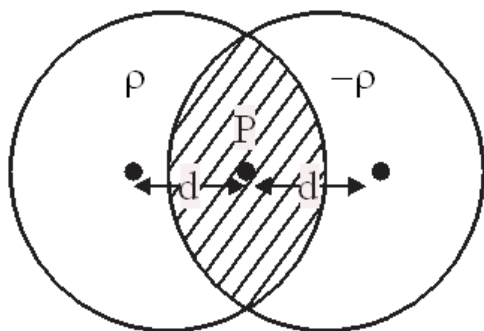
$$|a_i| = \frac{2m^3 u_0^2}{f}$$

$u_0 \rightarrow$ velocity of object

$m \rightarrow$ magnification

$f \rightarrow$ focal length.

17. Ans (B)



$$E_p = \frac{\rho d}{3\epsilon_0} + \frac{\rho d}{3\epsilon_0} = \frac{2\rho d}{3\epsilon_0}$$

$\vec{E}$ due to shaded region will not be that as it is O net charge region.

So, E_p is electric field due to remaining portion of each sphere. Hence, due to one sphere it is

$$\frac{\rho d}{3\epsilon_0}$$

18. Ans (A)

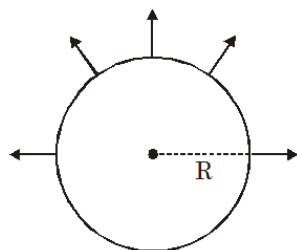
$$\frac{q_0^2}{2 \times 2c} = \frac{2(q_0/3)^2}{2 \times 2C} + \frac{(q_0/3)^2}{2 \times C} + \frac{1}{2} Li^2$$

$$\frac{q_0^2}{6C} = Li^2$$

$$i = \frac{q_0}{\sqrt{6LC}}$$

19. Ans (D)

In equilibrium of upper half of loop $2kx = 4RS$



$$x = \frac{2SR}{k} \Rightarrow 2\pi R = \ell + \frac{2SR}{k}$$

$$R = \frac{k\ell}{2\pi k - 2S}$$

20. Ans (B)

$$F = \alpha \beta e^{\left(-\frac{x}{\alpha k}\right)} \Rightarrow \frac{x}{\alpha} = [ML^2T^{-2}]$$

$$\alpha = [M^{-1}L^{-1}T^2]$$

$$\Rightarrow [MLT^{-2}] = [M^{-1}L^{-1}T^2] [\beta]^{-1}$$

$$\beta = [M^{-2}L^{-2}T^4]$$

PART-1 : PHYSICS

SECTION-II

1. Ans (215)

$$\text{Least count} = \frac{0.25}{5 \times 100} \text{ cm} = 5 \times 10^{-4} \text{ cm}$$

$$\text{Reading} = 4 \times 0.05 \text{ cm} + 30 \times 5 \times 10^{-4} \text{ cm} \\ = (0.2 + 0.0150) \text{ cm} = 2.15 \text{ mm}$$

2. Ans (4)

$$I = xH \text{ and } x = \frac{C}{T} \text{ (according to curie law)}$$

Magnetization (T)

$$\propto \frac{\text{Magnetic Induction (B)}}{\text{Absolute temperature (T)}} \Rightarrow \frac{I_2}{I_1} = \frac{B_2 T_1}{B_1 T_2}$$

3. Ans (5)

$$A = \frac{A_0}{\sqrt{e}} = A_0 e^{-\pi t}$$

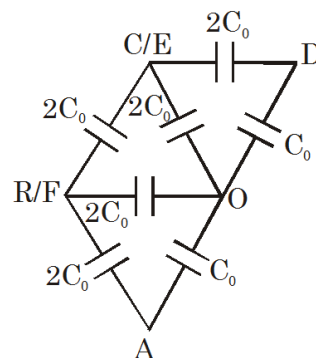
$$\frac{1}{2} = \pi t = \frac{b}{2m} t$$

$$t = \frac{m}{b}$$

4. Ans (20)

Using the parallel axis symmetry it can be reduced to the following circuit.

Solving this circuit will give us equivalent capacitance between A and O as $\left(\frac{20}{9}\right) C_0$

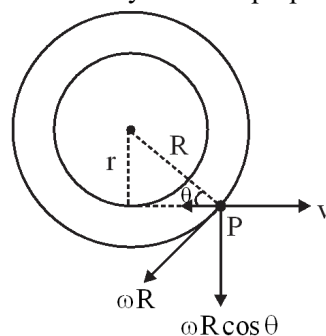


5. Ans (3)

Horizontal velocity of point

$$= v - \omega R \sin \theta = \omega r - \omega R \left(\frac{r}{R}\right) = 0$$

So velocity must be perpendicular.



6. **Ans (4)**

$$13.6z^2 \left[\frac{1}{1^2} - \frac{1}{(2n)^2} \right] = 204$$

$$13.6z^2 \left[\frac{1}{n^2} - \frac{1}{(2n)^2} \right] = 40.8$$

solving, $n = 2, z = 4$

7. **Ans (20)**

The weight of the chain is balanced by the vertical components of the reaction forces at the two suspension points. The magnitude of each of these components must be $F_0 = \frac{\rho L g}{2}$, where ρ is the mass of the chain per unit length, and $L = 40$ cm. If the chain made an angle of θ with the vertical at a suspension point, the horizontal component of the chain's tension would be $F_0 \tan \theta$. Because, in practice, $\theta = 45^\circ$, the horizontal component is the same as the vertical one; it is also constant along the chain, since there are horizontal forces acting on the chain only at its ends.

At the chain's lowest point, the tension is purely horizontal, and, as just shown, its magnitude is F_0 . So, around the lowest point, consider a small piece of the chain that subtends an angle 2θ at the centre of the osculating circle, whose radius is the value r_1 we seek. The length of the piece is $2\theta r_1$ and it is pulled down by a gravitational force $2\theta r_1 \rho g$. This force is balanced by the upward net force of $2F_0 \sin \theta$ due to the tension in the chain. In the limit of small angles, when $\sin \theta \approx \theta$, the equilibrium equation

$$2\theta r_1 \rho g = 2F_0 \sin \theta$$

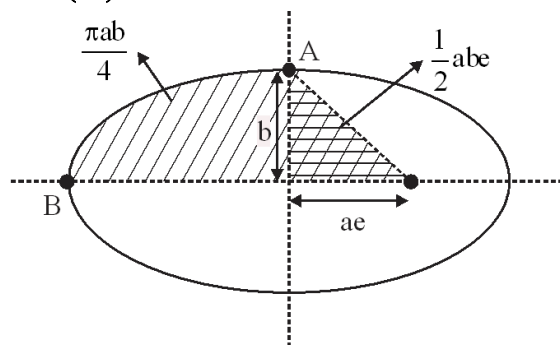
leads to

$$F_0 = \rho g r_1$$

Since $F_0 = \frac{\rho L g}{2}$, we have the simple result that

$$r_1 = \frac{L}{2} = 20 \text{ cm.}$$

8. **Ans (6)**



$$t = \frac{T}{\pi ab} \left(\frac{\pi ab}{4} + \frac{1}{2}abe \right)$$

$$\Rightarrow t = T \left(\frac{1}{4} + \frac{e}{2\pi} \right)$$

$$t = 18 \left(\frac{1}{4} + \frac{1}{8} \right) \Rightarrow t = 6 \text{ months}$$

9. **Ans (2)**

$$(8\pi \text{ cm}^2 \times 1.5 \text{ m/min}) = 40 \times v \times \frac{\pi (1 \text{ mm})^2}{4}$$

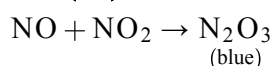
PART-2 : CHEMISTRY

SECTION-I

1. **Ans (A)**

A-B interaction is weaker compared to A-A or B-B interaction

2. **Ans (C)**



3. **Ans (D)**

HgCl_2 is water soluble

4. **Ans (D)**

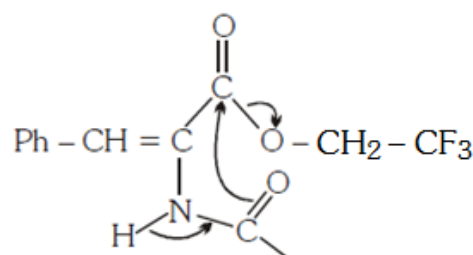
$$k_I = Ae^{-\frac{E_a}{RT}} \quad \dots(i)$$

$$k_{II} = Ae^{-\frac{2E_a}{RT}} \quad \dots(ii)$$

$$k'_I = Ae^{-\frac{E_a}{RT}} \quad \dots(iii)$$

$$k'_{II} = Ae^{-\frac{2E_a}{RT}} \quad \dots(iv)$$

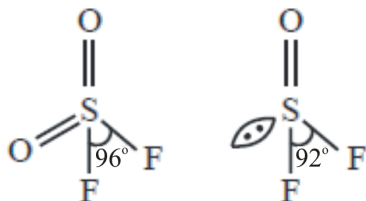
5. **Ans (B)**



6. **Ans (B)**

pH = 2.95 is isoelectric point

7. **Ans (C)**



8. **Ans (D)**

$\Delta G = \text{maximum electrical work} = -nFE$

So, a cell having maximum +ve E will do maximum work.

$$\text{Also, } E = \frac{0 - 0.059}{n} \frac{\log[H^+]_{\text{anode}}}{[H^+]_{\text{cathode}}}$$

Also, $\frac{[H^+]_{\text{anode}}}{[H^+]_{\text{cathode}}}$ is clearly minimum for option (D)

10. **Ans (A)**

A is more stable alkene due to resonance

13. **Ans (A)**

The orbital is 2s

* no of radial node = $n - l - 1 = 2 - 0 - 1 = 1$

* Probability density Ψ^2

* Since Ψ contain no θ or ϕ terms – probability of finding electron do not depend upon direction

* Also since at $r = 0$,

$$\psi = \left(\frac{1}{4\sqrt{2}} \right) \left(\frac{1}{a_0} \right)^{3/2} \cdot (2) ; \Psi^2 \text{ is non zero at } r = 0,$$

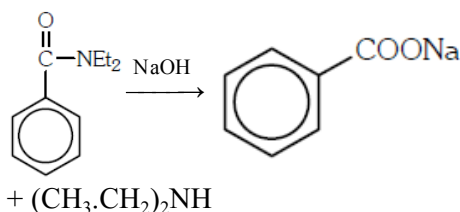
* $\Psi = 0$ when the term $\left(2 - \frac{r}{a_0} \right)$ vanishes

$$\text{putting } 2 - \frac{r}{a_0} = 0$$

$$\Rightarrow r = 2a_0$$

at $r = 2a_0$, $\Psi = 0$: Corresponds to radial node.

14. **Ans (D)**



15. **Ans (A)**

$$\Delta G = VdP - SdT$$

$$dT = 0$$

$$\therefore \Delta G = VdP = \int_{P_1}^{P_2} \frac{nRT}{P} dP$$

$$= nRT \ln \frac{P_2}{P_1} = +W = 1200 \text{ Cal.}$$

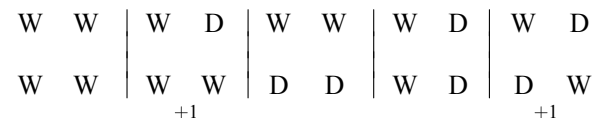
17. **Ans (B)**



$$1 - x \quad 2x$$

$$K_p = \frac{\left(\frac{2x}{1+x} \right)^2}{\left(\frac{1-x}{1+x} \right)} \cdot P_T$$

18. **Ans (C)**



19. **Ans (B)**

Complete octet and (–) charge on more EN element

20. **Ans (A)**

$$\text{Milli moles of } H^+ = 2.5 + 5.0 = 7.5$$

$$\text{Milli moles of } OH^- = 10$$

$$\text{Milli moles of } OH^- \text{ left after mixing} = 2.5$$

$$\text{Total volume} = 100 \text{ ml}$$

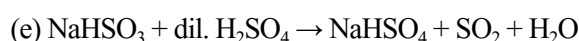
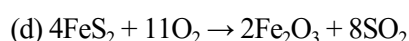
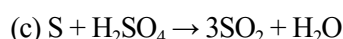
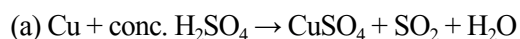
$$\text{Conc. of } OH^- = \frac{2.5}{100} = 2.5 \times 10^{-2}$$

$$p^{OH} = 2 - \log 2.5 = 2 - 0.7 + 0.3 = 1.6$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (4)**



2. **Ans (2)**

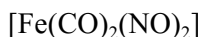
$$\frac{A_1}{A_2} = \frac{\lambda_1 N_1}{\lambda_2 N_2}$$

$$\frac{(N_0 e^{-\frac{\lambda_1 t}{2}} \times 2)}{(N_0 e^{-\frac{\lambda_2 t}{4}} \times 2)} = \frac{\frac{\lambda_2}{2} \times 2}{\frac{\lambda_1}{4}} = \sqrt{2}$$

3. **Ans (0)**



EAN = 26 - 0 + 10 = 36



EAN ⇒ 26 - 0 + 2 × 2 + 2 × 3 = 36

Difference ⇒ 0

4. **Ans (4)**

Number of moles of cation vacancy

$$= \frac{6.02 \times 10^{21}}{6.02 \times 10^{23}} = 10^{-2} \text{ mole}$$

Number of moles of M²⁺ ions = 10⁻² mole

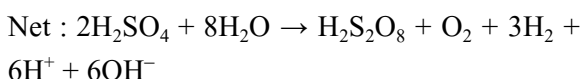
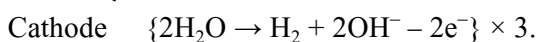
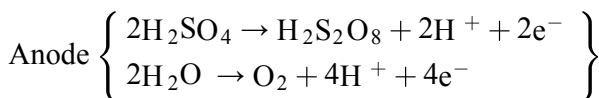
Number of moles of missing y⁻ ions = 2 × 10⁻² mole %

decreases in density

$$= \frac{30 \times 10^{-2} - 20 \times 10^{-2}}{25} \times 100\% = 0.4\% = x\%$$

⇒ 10x = 4

5. **Ans (3)**



Hence ratio of n_{O₂} and n_{H₂} is 1 : 3.

6. **Ans (5)**

(i), (ii), (vi), (vii), (ix) can evolve H₂ on reaction with Na metal.

7. **Ans (25)**

Energy of one photon = $\frac{12400}{3100} = 4\text{eV}$

= 4 × 96 = 384 kJ mol⁻¹

∴ % of energy converted to K.E.

$$= \frac{384 - 288}{384} = \frac{96}{384} \times 100 = 25\%$$

8. **Ans (224)**

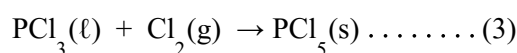
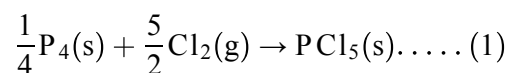
$$\Delta S = nR \ln \frac{P_1}{P_2} = 1 \times 8 \ln \frac{2}{4} = -5.6 \text{ JK}^{-1}$$

$$\Delta S_{\text{surr}} = \frac{-Q_{\text{sys}}}{T} = \frac{W_{\text{irr}}}{T} = \frac{1.2 \times W_{\text{rev}}}{300}$$

$$= \frac{1.2 \times 350 \times 5.6}{300} = 7.84 \text{ JK}^{-1}$$

$$\therefore \Delta S_{\text{total}} = 7.84 - 5.6 = 2.24 \text{ JK}^{-1}$$

9. **Ans (5)**



On applying thermodynamic operation

$$(1) = (3) + \frac{1}{2}(2)$$

$$= -200 + \frac{1}{2}(-600)$$

$$= -500 \text{ kJ}$$

$$= -5 \times 10^2 \text{ kJ}$$

10. **Ans (12)**

Maximum buffer capacity is obtained when pH = pK_{a1}

and when pH = pK_{a2}

pH at second equivalence point

$$= 7 + \frac{10}{2} + \frac{1}{2} \log 1 = 12$$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (C)**

Divide by x⁶ in numerator and denominator

$$\Rightarrow \int \frac{\left(\frac{1}{x^2} + \frac{3}{x^4} + \frac{5}{x^6} \right) dx}{\sqrt{1 - \frac{1}{x} - \frac{1}{x^3} - \frac{1}{x^5}}}$$

Now put $1 - \frac{1}{x} - \frac{1}{x^3} - \frac{1}{x^5} = t^2$

$$\Rightarrow \int \frac{2t dt}{t} \Rightarrow 2 \sqrt{1 - \frac{1}{x} - \frac{1}{x^3} - \frac{1}{x^5}} + C$$

2. **Ans (A)**

$$P(R) = \frac{2}{3}, P(G) = \frac{1}{3}$$

$$P\left(\frac{\text{Accident}}{R}\right) = \frac{1}{5}, P\left(\frac{\text{Accident}}{G}\right) = \frac{1}{8}$$

$$P\left(\frac{R}{\bar{A}}\right) = \frac{P(R \cap \bar{A})}{P(\bar{A})}$$

$$= \frac{P(R) \cdot P\left(\frac{\bar{A}}{R}\right)}{P(R)P\left(\frac{\bar{A}}{R}\right) + P(G) \cdot P\left(\frac{\bar{A}}{G}\right)}$$

$$= \frac{\frac{2}{3} \cdot \frac{4}{5}}{\frac{2}{3} \cdot \frac{4}{5} + \frac{1}{3} \cdot \frac{7}{8}} = \frac{64}{99}$$

3. **Ans (C)**

$$g(x) = px^2 + qx + r = 0 \text{ has roots } \alpha, \beta$$

$$\Rightarrow g(x) = p(x - \alpha)(x - \beta)$$

$$\text{Let } I = \int_{\alpha}^{\beta} \frac{e^{f\left(\frac{g(x)}{x-\alpha}\right)}}{e^{f\left(\frac{g(x)}{x-\alpha}\right)} + e^{f\left(\frac{g(x)}{x-\beta}\right)}} dx$$

$$= \int_{\alpha}^{\beta} \frac{e^{f(p(x-\beta))}}{e^{f(p(x-\beta))} + e^{f(p(x-\alpha))}} dx \quad \dots(i)$$

Apply king's rule

$$I = \int_{\alpha}^{\beta} \frac{e^{f(p(\alpha-x))}}{e^{f(p(\alpha-x))} + e^{f(p(\beta-x))}} dx \quad \dots(ii)$$

(replace $x \rightarrow a + b - x$)

$\therefore f$ is even

$$\Rightarrow \therefore f(p(\alpha - x)) = f(p(x - \alpha))$$

$$\& f(p(\beta - x)) = f(p(x - \beta))$$

Now add (i) and (ii)

$$\Rightarrow 2I = \int_{\alpha}^{\beta} dx \Rightarrow I = \frac{\beta - \alpha}{2}$$

$$\Rightarrow I = \frac{\sqrt{q^2 - 4pr}}{2|p|}$$

4. **Ans (B)**

$$\sum_{r=1}^{15} \frac{(r+3-3) \cdot 3^r}{(r+3)!} = \sum_{r=1}^{15} \frac{3^r}{(r+2)!} - \frac{3^{r+1}}{(r+3)!}$$

$$= \frac{3}{3!} - \frac{3^{16}}{18!}$$

5. **Ans (B)**

$$I = \int_0^1 \frac{\sin x}{x^{1/4}} dx < \int_0^1 \frac{x \cdot dx}{x^{1/4}}$$

$$\Rightarrow I < \frac{4}{7} \quad (\because \sin x < x \forall x \in (0, 1))$$

$$J = \int_0^1 \frac{\cos x}{x^{1/4}} \cdot dx < \int_0^1 \frac{1}{x^{1/4}} dx$$

$$\Rightarrow J < \frac{4}{3} \quad (\because \cos x < 1 \forall x \in (0, 1))$$

6. **Ans (B)**

$$(\vec{p} \cdot \vec{p})(\vec{x} - \vec{q}) - (\vec{p} \cdot (\vec{x} - \vec{q}))\vec{p}$$

$$+ (\vec{q} \cdot \vec{q})(\vec{x} - \vec{r}) - (\vec{q} \cdot (\vec{x} - \vec{r}))\vec{q}$$

$$+ (\vec{r} \cdot \vec{r})(\vec{x} - \vec{p}) - (\vec{r} \cdot (\vec{x} - \vec{p}))\vec{r} = 0$$

$$\because |\vec{p}| = |\vec{q}| = |\vec{r}| = \lambda (\text{let})$$

$$\Rightarrow \lambda^2 (3\vec{x} - (\vec{p} + \vec{q} + \vec{r}))$$

$$= (\vec{p} \cdot \vec{x})\vec{p} + (\vec{q} \cdot \vec{x})\vec{q} + (\vec{r} \cdot \vec{x})\vec{r}$$

$$\Rightarrow 3\vec{x} - (\vec{p} + \vec{q} + \vec{r}) = \vec{x}$$

7. **Ans (A)**

Let

$$L = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n^2}\right)^{\frac{2}{n^2}} \left(1 + \frac{2^2}{n^2}\right)^{\frac{4}{n^2}} \left(1 + \frac{3^2}{n^2}\right)^{\frac{6}{n^2}} \dots \left(1 + \frac{n^2}{n^2}\right)^{\frac{2n}{n^2}}$$

$$\Rightarrow \ln L = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{2r}{n^2} \ln \left(1 + \frac{r^2}{n^2}\right)$$

$$\Rightarrow \ln L = \int_0^1 2x \cdot dx \cdot \ln(1 + x^2)$$

$$\Rightarrow \ln L = \int_1^2 \ln t \cdot dt = (t \ln t - t)_1^2$$

$$\Rightarrow \ln L = 2 \ln 2 - 1 \Rightarrow L = \frac{4}{e}$$

8. **Ans (D)**

$$f(x) = [x] + 2 + [x]^2 - 8$$

$$f(x) = ([x] + 3)([x] - 2)$$

Clearly $f(x) = 0$ for $x \in [2, 3)$ many one

Also output of $f(x)$ is always an integer.

$\Rightarrow f(x)$ is into

9. **Ans (B)**

$$\frac{\cot B}{\cot C + \cot A} = \frac{\frac{\cos B}{\sin B}}{\left(\frac{\sin(A+C)}{\sin A \sin C}\right)}$$

$$= \cos B \cdot \frac{\sin A \sin C}{\sin^2 B}$$

$$= \frac{a^2 + c^2 - b^2}{2ac} \cdot \frac{ac}{b^2} = \frac{1}{2}$$

10. **Ans (C)**

If line lies on plane

$$\Rightarrow (3\hat{i} + 5\hat{j} + 2\hat{k}) \cdot (\alpha\hat{i} + 2\hat{j} + \hat{k}) = 0$$

$$\Rightarrow 3\alpha + 10 + 2 = 0 \Rightarrow \alpha = -4$$

Point (2, -1, 2) on line must satisfy plane eqn.

$$\Rightarrow 2\alpha - 2 + 2 - \beta = 0$$

$$\Rightarrow \beta = -8$$

11. **Ans (B)**

Eqn. of tangent of hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ is

$$y = mx \pm \sqrt{9m^2 - 4} \quad \dots(1)$$

Eqn. of tangent of circle

$$y = mx \pm \sqrt{5m^2 + 5} \quad \dots(2)$$

Compare (1) and (2)

$$\Rightarrow 9m^2 - 4 = 5m^2 + 5 \Rightarrow m = \frac{3}{2}$$

required tangent

$$3x - 2y \pm \sqrt{65} = 0$$

12. **Ans (B)**

$$\text{Let } \frac{z-1}{e^{i\frac{\pi}{3}}} = t$$

$$\Rightarrow t + \frac{1}{t} = \text{purely real}$$

$$\Rightarrow t + \frac{1}{t} = \bar{t} + \frac{1}{\bar{t}} \Rightarrow (t - \bar{t}) \left(1 - \frac{1}{t\bar{t}}\right) = 0$$

$$\Rightarrow t = \bar{t} \quad (\because |t| \neq 1 \because |z-1| \neq 1)$$

$$\Rightarrow t = \text{purely real}$$

$$\Rightarrow \frac{x+y-1}{e^{i\frac{\pi}{3}}} = \text{purely real}$$

$$\Rightarrow ((x-1) + iy) \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3}\right)$$

= purely real

$$\Rightarrow -(x-1) \cdot \frac{\sqrt{3}}{2} + y \cdot \frac{1}{2} = 0$$

$$\Rightarrow -\sqrt{3}x + y + \sqrt{3} = 0$$

Slope of line = $\sqrt{3}$

13. **Ans (D)**

$$\text{Let } h(x) = \int_0^{x^3} f(t) dt$$

$$\Rightarrow h'(x) = 3x^2 \cdot f(x^3)$$

Apply LMVT on h(x)

$$\text{for } (0, 1) h'(\alpha) = \frac{h(1) - h(0)}{1 - 0} \quad \dots(1)$$

$$\text{for } (1, 2) h'(\beta) = \frac{h(2) - h(1)}{2 - 1} \quad \dots(2)$$

$$\text{for } (2, 3) h'(\gamma) = \frac{h(3) - h(2)}{3 - 2} \quad \dots(3)$$

Add (1), (2) and (3)

$$\Rightarrow h(3) = h'(\alpha) + h'(\beta) + h'(\gamma)$$

14. **Ans (D)**

Required ways = ${}^{16}C_5$

$$\therefore p - q = 16 - 5 \text{ or } 16 - 11$$

$$= 11 \text{ or } 5$$

15. **Ans (C)**

$$p \Leftrightarrow q = (p \Rightarrow q) \wedge (q \Rightarrow p)$$

Also it's fact that

$$p \Rightarrow q \equiv \sim p \vee q$$

16. **Ans (A)**

Differentiate both sides w.r.t. x

$$x^2 \cdot y(x) = 3x^2 \cdot y(x) + x^3 \cdot y'(x)$$

$$\Rightarrow x \frac{dy}{dx} + 2y = 0$$

$$\Rightarrow \frac{dy}{y} + \frac{2dx}{x} = 0$$

$$\Rightarrow \ln y + 2 \ln x = C$$

$$\Rightarrow x^2 y = C'$$

17. **Ans (D)**

$$\sum_{r=1}^{n-1} \tan^{-1} \left(\frac{2d}{1 + a_{2r-1} \cdot a_{2r+1}} \right)$$

$$\Rightarrow \sum_{r=1}^{n-1} \tan^{-1} \left(\frac{a_{2r+1} - a_{2r-1}}{1 + a_{2r-1} \cdot a_{2r+1}} \right)$$

$$\Rightarrow \sum_{r=1}^{n-1} \tan^{-1} (a_{2r+1}) - \tan^{-1} (a_{2r-1})$$

$$\Rightarrow \tan^{-1} (a_{2n-1}) - \tan^{-1} (a_1)$$

18. **Ans (D)**

for $x > 0$

$$x < \sqrt{2}x < \sqrt{3}x$$

$$\Rightarrow f(x) > f(\sqrt{2}x) > f(\sqrt{3}x)$$

$$\Rightarrow 1 > \frac{f(\sqrt{2}x)}{f(x)} > \frac{f(\sqrt{3}x)}{f(x)}$$

Apply limit $x \rightarrow \infty$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{f(\sqrt{2}x)}{f(x)} = 1$$

(by concept of sandwich theorem)

19. **Ans (C)**

$$A = \begin{bmatrix} 1 & 0 \\ \frac{1}{3} & 1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 0 \\ \frac{1}{3} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{1}{3} & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0 & 0+0 \\ \frac{1}{3}+\frac{1}{3} & 0+1 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 1 & 0 \\ \frac{2}{3} & 1 \end{bmatrix}$$

$$\text{Now } A^3 = \begin{bmatrix} 1 & 0 \\ \frac{2}{3} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{1}{3} & 1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 1 & 0 \\ \frac{3}{3} & 1 \end{bmatrix} \Rightarrow A^{48} = \begin{bmatrix} 1 & 0 \\ \frac{48}{3} & 1 \end{bmatrix}$$

20. **Ans (C)**

$$\tan \alpha + \tan \beta = -p$$

$$\text{and } \tan \alpha \cdot \tan \beta = q$$

$$\Rightarrow \tan(\alpha + \beta) = \frac{-p}{1-q}$$

$$\sin^2(\alpha + \beta) + p \sin(\alpha + \beta)$$

$$\cos(\alpha + \beta) + q \cos^2(\alpha + \beta)$$

$$= \cos^2(\alpha + \beta)$$

$$[\tan^2(\alpha + \beta) + p \tan(\alpha + \beta) + q]$$

$$= \frac{\frac{p^2}{(1-q)^2} - \frac{p \cdot p}{1-q} + q}{1 + \frac{p^2}{(1-q)^2}}$$

$$= \frac{p^2 - p^2(1-q) + q(1-q)^2}{(1-q)^2 + p^2} = q$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (4)**

$$(\alpha^2 + 2\beta^2 - 5)x^2 + x(2\alpha^2 - \beta^2) +$$

$$(\alpha^2 + \beta^2 - 3) = 0$$

$$\forall x \in \mathbb{R}$$

$$\therefore \alpha^2 + 2\beta^2 - 5 = 0 \text{ \& } 2\alpha^2 - \beta^2 = 0$$

$$\text{ \& } \alpha^2 + \beta^2 = 3$$

$$\therefore \alpha^2 + 2.2\alpha^2 = 5 \text{ \& } \alpha^2 + 2\alpha^2 = 3$$

$$\Rightarrow \alpha^2 = 1 \text{ \& } \beta^2 = 2$$

$$\Rightarrow (\alpha, \beta) = (\pm 1, \pm \sqrt{2})$$

2. **Ans (3)**

ω is cube root of unity

$$|x|^2 = (a + b + c)(\bar{a} + \bar{b} + \bar{c})$$

$$= |a|^2 + |b|^2 + |c|^2 + a\bar{b} + \bar{a}b$$

$$+ b\bar{c} + \bar{b}c + c\bar{a} + \bar{c}a \dots (1)$$

$$|y|^2 = y\bar{y} = (a + b\omega + c\omega^2)(\bar{a} + \bar{b}\omega^2 + \bar{c}\omega)$$

$$= |a|^2 + |b|^2 + |c|^2 + a\bar{b}\omega^2 + \bar{a}b\omega$$

$$+ b\bar{c}\omega^2 + \bar{b}c\omega + c\bar{a}\omega^2 + \bar{c}a\omega \dots (2)$$

Similarly

$$|z|^2 = z\bar{z} = (a + b\omega^2 + c\omega)(\bar{a} + \bar{b}\omega + \bar{c}\omega^2) \dots (3)$$

$$\text{Now add (1), (2), (3) and use } 1 + \omega + \omega^2 = 0$$

$$\text{to get } |z|^2 + |y|^2 + |x|^2 = 3(|a|^2 + |b|^2 + |c|^2)$$

3. **Ans (2)**

$$I = \int_0^2 \frac{17 - 2(x-1)^2}{e^{6(x-1)} + 1} dx \dots (1)$$

Apply king's rule

$$I = \int_0^2 \frac{17 - 2(1-x)^2}{e^{6(1-x)} + 1} dx \dots (2)$$

Add (1) and (2)

$$2I = \int_0^2 17 - 2(x-1)^2 dx$$

$$\Rightarrow 2I = \left(17x - \frac{2(x-1)^3}{3} \right)_0^2$$

$$\Rightarrow 2I = 34 - \frac{2}{3} - \frac{2}{3} \Rightarrow I = \frac{49}{3}$$

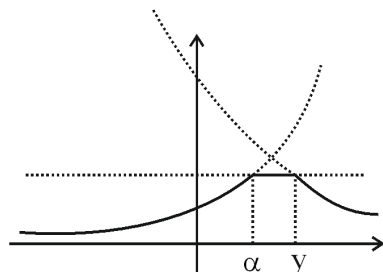
4. Ans (2)

$$A = {}^{22}C_{11} = \frac{22}{11} \cdot {}^{21}C_{10}$$

$$B = {}^{21}C_{11}$$

$$\therefore \frac{A}{B} = 2$$

5. Ans (2)



$$e^\alpha = \frac{3}{2} \Rightarrow \alpha = \ln \frac{3}{2}$$

$$e^\beta = 1 + e^{-\beta} \Rightarrow e^\beta = \frac{1 + \sqrt{5}}{2}$$

$$\Rightarrow \beta = \ln \left(\frac{1 + \sqrt{5}}{2} \right)$$

$$\& e^{-y} = \frac{1}{2} \Rightarrow y = \ln 2$$

so at 2 points $x = \alpha, y$ curve has sharp corners.

6. Ans (1)

Apply L'Hopital rule

$$\lim_{x \rightarrow 0} \frac{\sec^3(x^4) \cdot 2x}{x \cos x + \sin x}$$

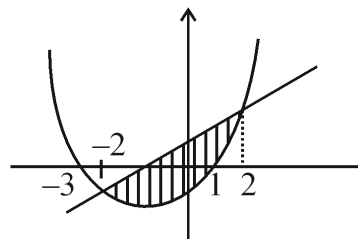
$$\lim_{x \rightarrow 0} \frac{\sec^3(x^4) \cdot 2}{\cos x + \frac{\sin x}{x}} \Rightarrow \frac{2 \times 1}{1 + 1} \Rightarrow 1$$

7. Ans (32)

$$y = x^2 + 2x - 3 \text{ and } y = 2x + 1$$

both curves intersect when $x^2 + 2x - 3 = 2x + 1$

$$\Rightarrow x = \pm 2$$



$$\begin{aligned} \text{Area} &= - \int_{-2}^2 (x^2 + 2x - 3 - (2x + 1)) dx \\ &= - \left(\frac{x^3}{3} - 4x \right)_{-2}^2 \\ &= \frac{32}{3} \end{aligned}$$

8. Ans (1)

If first observation is x_i then second observation is $x_i + 50$

$$\therefore \sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N}$$

$$\text{so, } \sigma_A^2 = \sigma_B^2$$

9. Ans (33)

$$\left(\sqrt{3} + 5^{1/8} \right)^{256} = \sum_{r=0}^{256} {}^{256}C_r \cdot (\sqrt{3})^{256-r} \cdot (5)^{r/8}$$

so for integral terms $r = 0, 8, 16, \dots, 256$ total 33 terms

10. Ans (0)

$$x^2 \cdot f_1(x) - 2f_1\left(\frac{1}{x}\right) = f_2(x) \quad \dots(1)$$

replace $x \rightarrow -x$ and then add with (1)

$$\Rightarrow 2x^2 f_1(x) - 4f_1\left(\frac{1}{x}\right) = 0$$

$$(\because f_1(-x) = f_1(x) \text{ \& } f_2(-x) = -f_2(x))$$

$$\therefore x^2 f_1(x) - 2f_1\left(\frac{1}{x}\right) = 0 \quad \dots(2)$$

Now replace $x \rightarrow \frac{1}{x}$ into (2)

$$\Rightarrow \frac{1}{x^2} f_1\left(\frac{1}{x}\right) - 2f_1(x) = 0 \quad \dots(3)$$

solve (2) and (3) to get $f_1(x) = 0 \Rightarrow f_1(3) = 0$